

Diabetes clinical trials across three start-date periods: a descriptive registry analysis

Abstract

Objective. Compare registered interventional diabetes research across start-date cohorts and identify practical sources of trial information and research datasets.

Methods. A cross-sectional extraction of ClinicalTrials.gov records used the condition query diabetes, interventional study type, and registered start dates from January 1, 2010 through October 5, 2026. All returned pages were retrieved, records were deduplicated by NCT identifier, and counts were grouped by registered start year. Actual, estimated, and untyped dates were retained and separately audited. This is an original descriptive registry analysis, not a systematic literature review or peer-reviewed publication.

Results. The extraction contained 15,087 unique records: 4,751 in the 2010–2015 cohort, 4,372 in 2016–2020, and 5,964 in the partial 2021–2026 cohort. Randomized allocation was recorded for 77.9%, 75.3%, and 75.6%, respectively. Registry results were posted for 28.2%, 21.2%, and 6.8%, respectively. Percentages were calculated using all records in each cohort as the denominator.[1]

Conclusion. Annual registered start counts were higher in several recent complete years than in the early years, but the analysis does not establish improved treatment efficacy or worse reporting compliance. It describes the retrieved registry set, not every diabetes trial worldwide. Estimated dates, missing date-type labels, incomplete follow-up, and broad condition-query matching constrain interpretation.[2, 3]

Research question and scope

The research question is whether registered trial activity, randomized design, completion status, and results availability differ between the requested start-date cohorts. Publication dates were not used to assign cohorts. The analysis includes all statuses and geographic locations returned by the query, and it does not impose age, sponsor, intervention-class, or phase restrictions.

The scope is deliberately broad: records returned by the diabetes condition query can concern prevention, complications, or related conditions rather than diabetes treatment alone. Type 1, type 2, gestational diabetes, and prevention studies were not manually adjudicated into separate subgroups. The downloadable condition and intervention fields support subsequent clinical screening. This is a query-defined dataset rather than an adjudicated census of diabetes trials.

Methods

Data source and retrieval

Records were obtained from the public ClinicalTrials.gov study endpoint: <https://clinicaltrials.gov/api/v2/studies>. The condition query was diabetes; the advanced filter was `AREA[StudyType] INTERVENTIONAL AND AREA[StartDate] RANGE[2010-01-01, 2026-10-05]`. Retrieval continued through the final page. One row was retained per NCT identifier.

The date cutoff applies to the registered start date. Status, enrollment, and results fields reflect the live records at retrieval; this is not a reconstruction of the registry as it appeared at each historical period. Later updates to older studies can change their current fields. The analysis cutoff is October 5, 2026, and the latest cohort is explicitly partial.

Variables and analysis

The trial-level CSV contains NCT identifier, title, start date and date type, year, cohort, current status, phase, enrollment count and enrollment type, allocation, lead sponsor, conditions, intervention names, results-posted indicator, last update date, and a public registry link. Missing values remain missing. Empty phase fields are labeled “Not applicable / not reported,” without assigning an inferred drug-development phase.

Counts were calculated by year and cohort. Randomized counts use the allocation value RANDOMIZED; completed counts use current status COMPLETED; results-posted counts use the registry's results indicator. These indicators are separate, not mutually exclusive. Results-posted does not mean that a treatment was successful. No treatment effects were pooled, no significance tests were performed, and no causal model was fitted.

Results

Registered trial activity

The cohort totals were 4,751, 4,372, and 5,964 records. Directly comparing these totals is misleading because the requested periods span unequal durations and the latest period is incomplete. Annual counts are therefore the more informative activity display.[1]

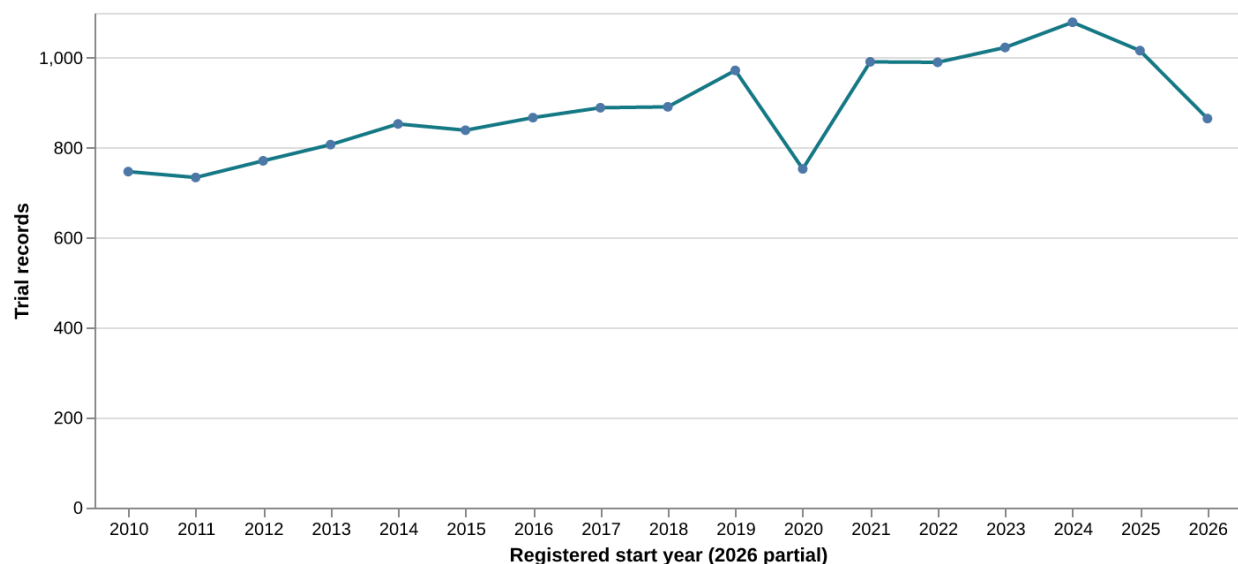


Figure 1. Counts include actual, estimated, and untyped start dates. The final year is partial; these are registry activity counts, not treatment effects.

Annual registered start counts increased from 747 in 2010 to 972 in 2019, fell to 753 in 2020, and then reached 1,079 in 2024. The 2025 count was 1,016. The partial 2026 count was 865 and should not be interpreted as evidence of a full-year decline. The dataset alone cannot explain the dip in 2020.[2]

Design, completion, and results availability

Randomized allocation appeared in 3,700 early-cohort records, 3,292 middle-cohort records, and 4,507 recent-cohort records. The corresponding proportions were 77.9%, 75.3%, and 75.6%. This suggests broadly similar recorded allocation patterns, not equivalence in trial quality or risk of bias.[1]

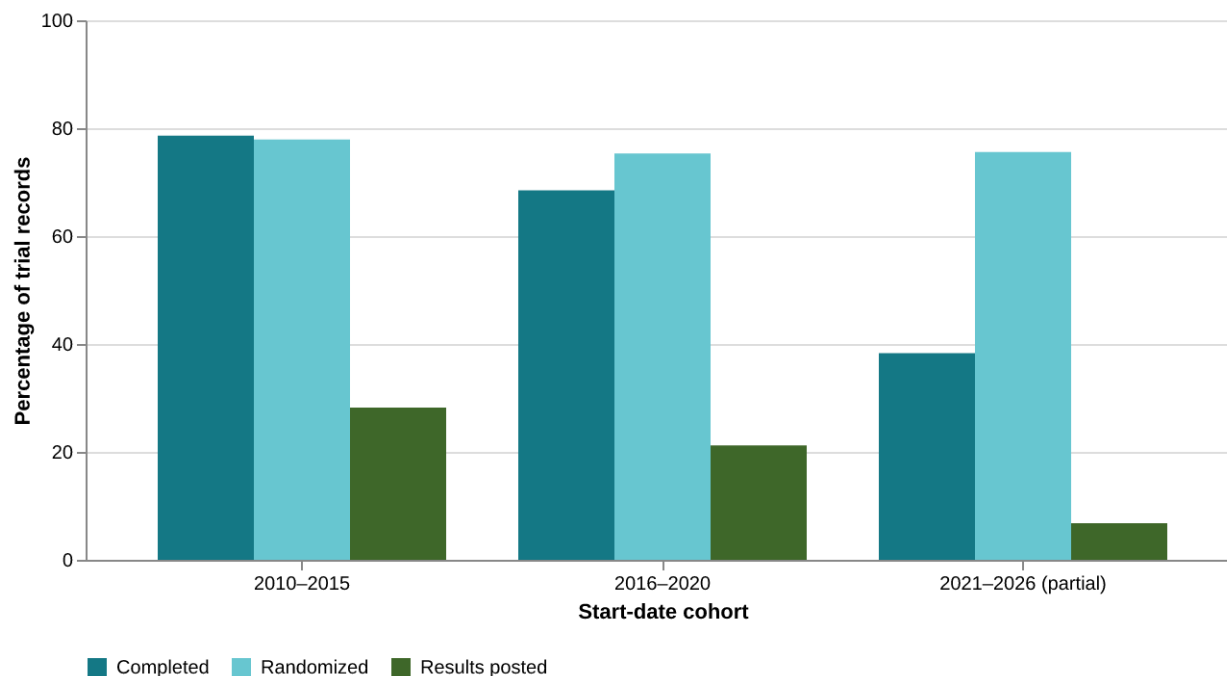


Figure 2. *Percent of each cohort with randomized allocation, completed status, or posted registry results. Unequal follow-up prevents interpreting these differences as reporting compliance.*

Completed status occurred in 3,734, 2,996, and 2,284 records, or 78.6%, 68.5%, and 38.3% of each cohort. Posted results appeared in 1,340, 927, and 408 records, or 28.2%, 21.2%, and 6.8%. These are cross-sectional availability measures, not reporting-compliance estimates. A valid compliance comparison would require eligibility for reporting, completion timing, applicable deadlines, and comparable follow-up.[1]

Start-date quality

The early cohort had 1,112 dates labeled actual, 9 estimated, and 3,630 without a date-type label. The middle cohort had 3,764 actual, 245 estimated, and 363 untyped dates. The recent cohort had 4,807 actual and 1,157 estimated dates, with no untyped dates. Thus, the main graphs include planned as well as actual starts; they must not be described as counts of confirmed trial initiations.[3]

Restricting to actual-labeled dates would disproportionately remove older records because of missing date-type labels. For this reason, an actual-only comparison is a sensitivity view, not automatically a less biased historical trend. The separate start-date-quality CSV preserves the relevant counts.[3]

Websites and access to datasets

The accompanying website directory lists key resources, not every clinical-research website on the internet. Registry records and participant-level research datasets serve different purposes and should not be conflated.

- **ClinicalTrials.gov:** <https://clinicaltrials.gov/> and its study-data endpoint above were the source for this analysis. The delivered dataset is trial-level registry metadata, not individual patient records.
- **WHO ICTRP:** <https://trialsearch.who.int/> provides access to data from multiple registry providers, links to original records, and groups multiple registrations concerning the same trial. WHO provides CSV and XML download guidance.[4, 5]
- **AACR:** <https://aact.ctti-clinicaltrials.org/> provides a relational representation of ClinicalTrials.gov protocol and results data. Its downloads include PostgreSQL snapshots and pipe-delimited files. An account is required for access and downloads under the stated access policy.[6, 7]
- **NIDDK Central Repository:** <https://repository.niddk.nih.gov/> offers study-data requests and lists clinical phenotype, laboratory, genetic, survey, and continuous glucose monitoring data types. Access requires an

approved request rather than unrestricted patient-data downloading.[8, 9]

- **Vivli:** <https://vivli.org/> supports discovery of NIDDK studies. For those studies, resources are requested through NIDDK and released after approval; approved packages may be used in a designated secure analysis environment.[9]
- **ClinicalStudyDataRequest:** <https://clinicalstudydatarequest.com/> facilitates sponsor-contributed patient-level data access with proposal review and privacy protections. Study availability and current sponsor conditions must be checked individually; the retrieved page's headline metrics were not used.[10]

Discussion and limitations

The annual pattern is consistent with greater recorded trial activity in several recent complete years than in the earliest years. However, registry counts do not measure clinical benefit, scientific novelty, or the amount of usable patient-level evidence. A trial registered for a diabetes-related complication also need not be a trial of glucose-lowering treatment. The broad, unadjudicated search scope should be preserved when reusing the graphs.[2]

The lower recent-cohort completion and results-posting proportions cannot establish deteriorating reporting. Recent trials have had less opportunity to finish and post results; some records also contain estimated starts. A reporting analysis should instead establish a common time since completion and identify records subject to reporting requirements. This dataset has not performed those adjudications.[1, 3]

Coverage is restricted to ClinicalTrials.gov query matches. Cross-registry expansion through WHO ICTRP would require deduplication of registrations concerning the same trial, rather than simply adding registry counts. No publication linkage, outcome extraction, participant-level analysis, diabetes-subtype validation, or formal risk-of-bias assessment was undertaken. WHO's grouping of multiple records highlights the importance of trial-level deduplication in an international extension.[4]

Conclusion

This analysis provides a reproducible descriptive starting point for comparing diabetes-related interventional research across the requested periods. Annual recorded activity was higher in several recent complete years, while randomized allocation proportions remained broadly similar. Differences in completion and posted results should be read as cohort maturity and availability patterns, not as treatment-effect or compliance findings. The next substantive step is clinical adjudication by diabetes subtype and intervention, followed by linkage to outcomes for a clearly defined clinical question.[1, 2]

Data availability

The supplied downloads include the trial-level registry CSV, annual counts, period-level counts, and start-date-quality counts. Public NCT links allow individual records to be checked. These files contain registry information and derived aggregate counts; no individual participant data are supplied. The website directory identifies separate request routes for participant-level datasets.[9, 10]

References

1. period_summary.csv
2. annual_counts.csv
3. start_date_quality.csv
4. ICTRP Search Portal. <https://trialsearch.who.int/>
5. Downloading records from the ICTRP database. <https://www.who.int/tools/clinical-trials-registry-platform/network/who-data-set/downloading-records-from-the-ictrp-database>

6. Aact - Clinical Trials Transformation Initiative. <https://aact.ctti-clinicaltrials.org/>
7. AACT Database | Clinical Trials Transformation Initiative. <https://aact.ctti-clinicaltrials.org/downloads>
8. NIDDK Central Repository. <https://repository.niddk.nih.gov/home>
9. NIDDK-CR R4R- a program of the National Institute ... <https://vivli.org/ourmember/niddk-cr-r4r-a-program-of-the-national-institute-of-diabetes-and-digestive-and-kidney-diseases-nih/>
10. Clinical Study Data Request. <https://clinicalstudydatarequest.com/>